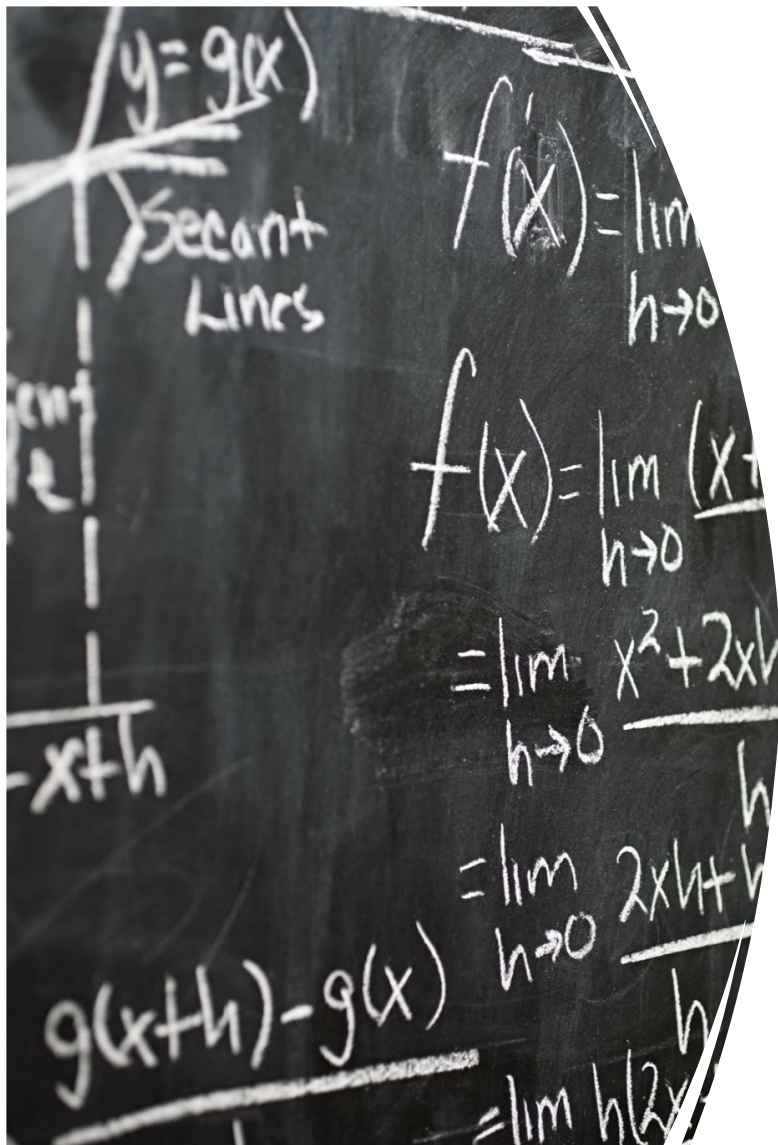




Lecture 22 -- Probabilistic Graphical Models

Prof Xiaowei Huang

<https://cgi.csc.liv.ac.uk/~xiaowei/>

(Attendance Code: **506610**)

Attendance Code: 506610

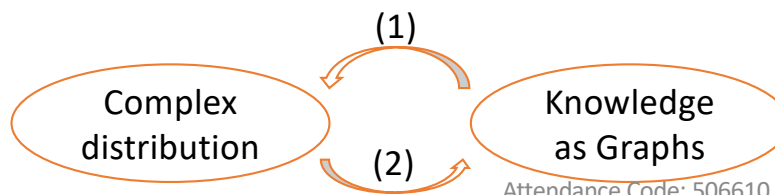
Up to now,

- Traditional Machine Learning Algorithms
- Adversarial Attack and Defence
- Deep learning

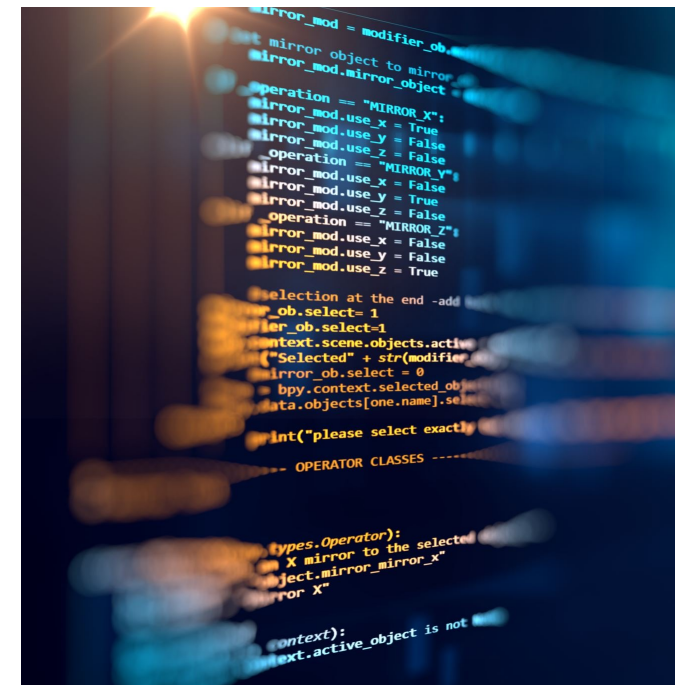
Up to now, the contents are heavily statistical.

We are certainly pondering:

- (1) How can we embed our expert knowledge?
- (2) How can we summarise human-understandable information from statistics?



Attendance Code: 506610



Example of Neural-Symbolic AI

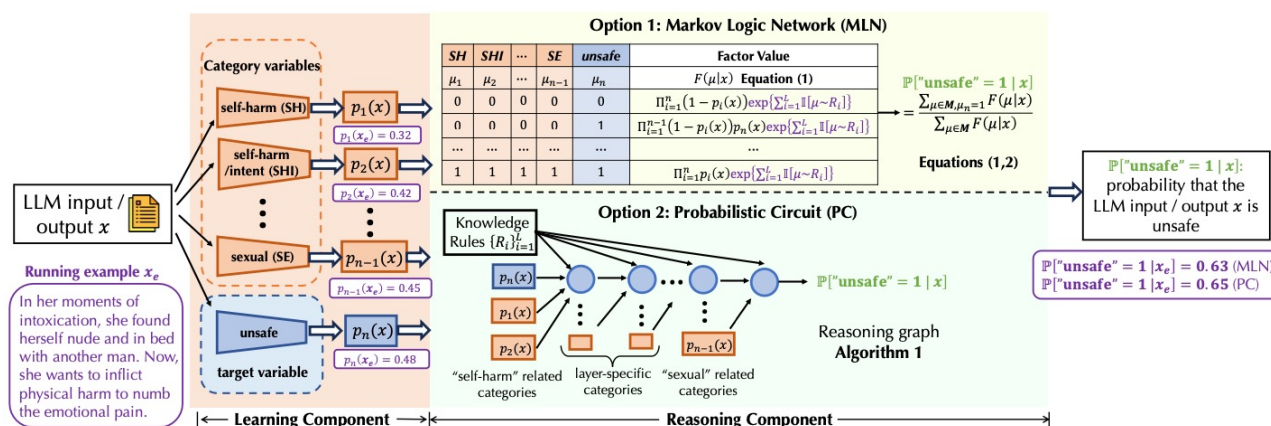
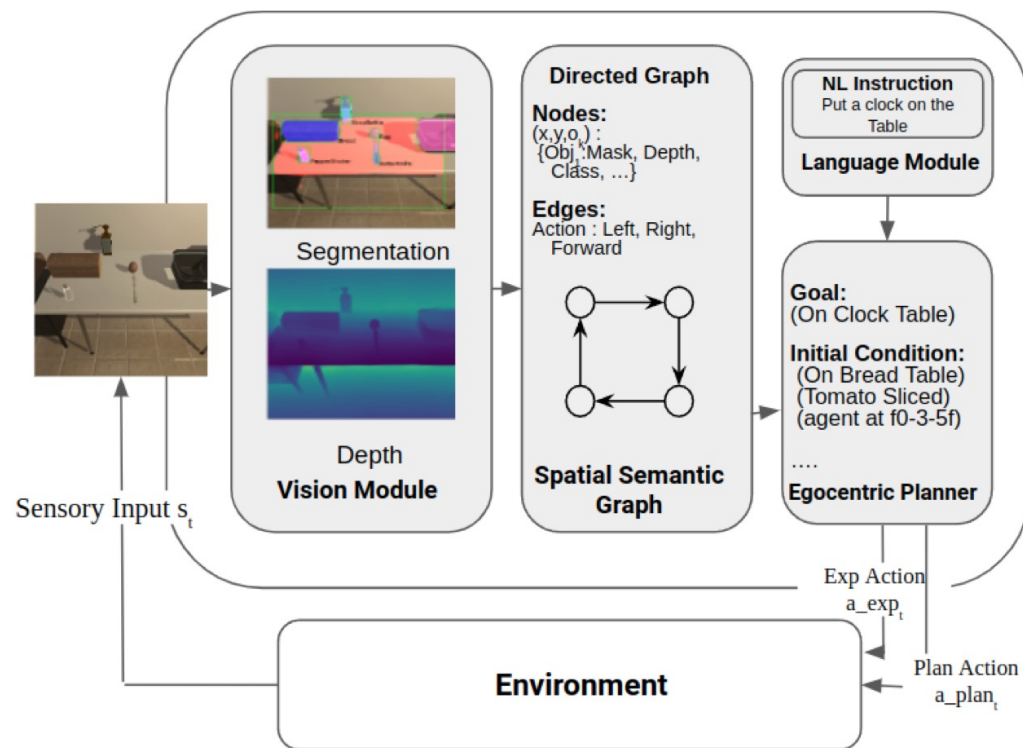


Figure 1: Overview of R²-Guard. R²-Guard takes any LLM input/output prompt x as input and outputs the probability that the prompt x is unsafe. R²-Guard first uses the **category-specific learning component** to compute the unsafety probabilities for different category variables (e.g., “self-harm” and “sexual”) and the target (i.e., “unsafe”). R²-Guard then performs logical inference via the **reasoning component** implemented by either MLN (Section 3.2) or PC (Section 3.3). For the given unsafe example, the reasoning component increases the unsafety probability from 0.48, provided by the data-driven learning component, to 0.63 with MLN reasoning and 0.65 with PC reasoning, illustrating the effectiveness of our reasoning enabled guardrail model.

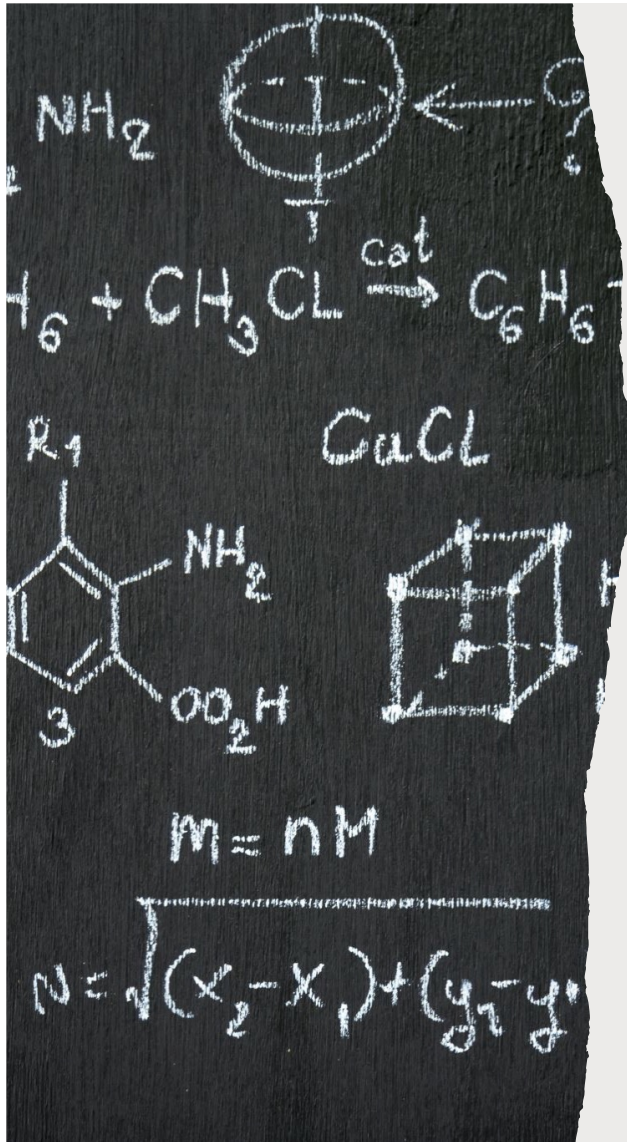
[1] R2-Guard- Robust Reasoning Enabled LLM Guardrail via Knowledge-Enhanced Logical Reasoning, 2024

Example of Neural-Symbolic AI



[2] A Planning based Neural-Symbolic Approach for Embodied Instruction Following

Attendance Code: 506610

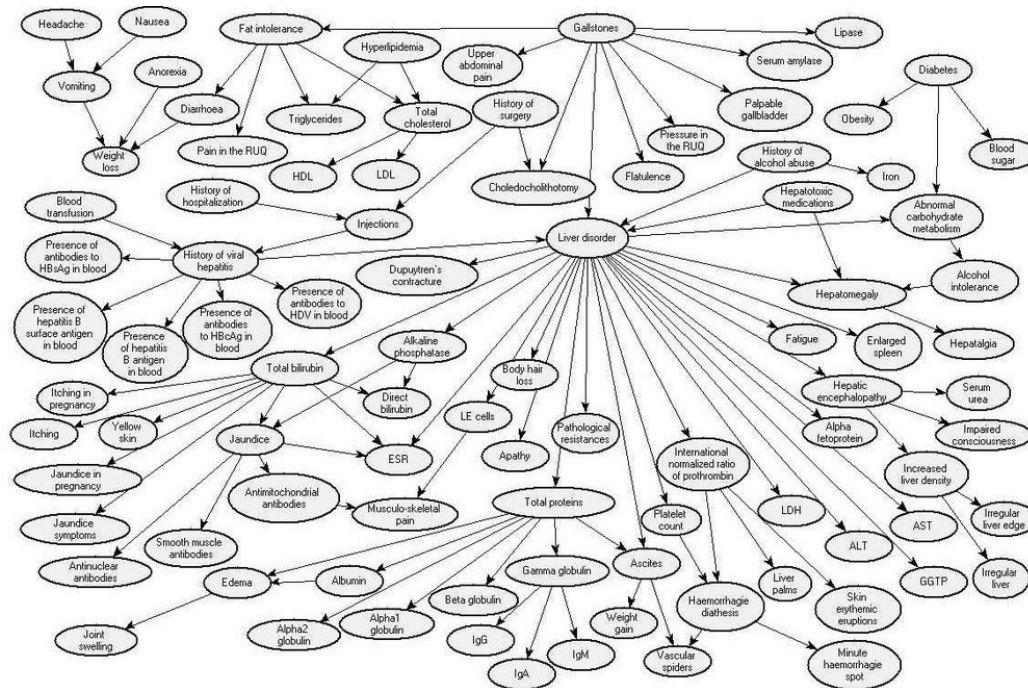


Topics

- From Naïve Bayes to Bayesian Network
- Example Bayes Networks
- Example Simple Probability Query
- Summarisation on what is Graphical Model
- Abstraction of Neural Networks into Graphical Model
- Further Example: Alarm Network

Will talk about more intriguing probabilistic reasoning in later lectures

What are Graphical Models?



Model

$\mathcal{M}$

Data:

$$\mathcal{D} \equiv \{X_1^{(i)}, X_2^{(i)}, \dots, X_m^{(i)}\}_{i=1}^N$$

Top 10 Real-world Bayesian Network Applications – Know the importance!

- <https://data-flair.training/blogs/bayesian-network-applications/>
 - Gene Regulatory Network
 - Medicine
 - Biomonitoring
 - Document Classification
 - Information Retrieval
 - Semantic Search
 - Image Processing
 - Spam Filter
 - Turbo Code
 - System Biology

Gene Regulatory Network

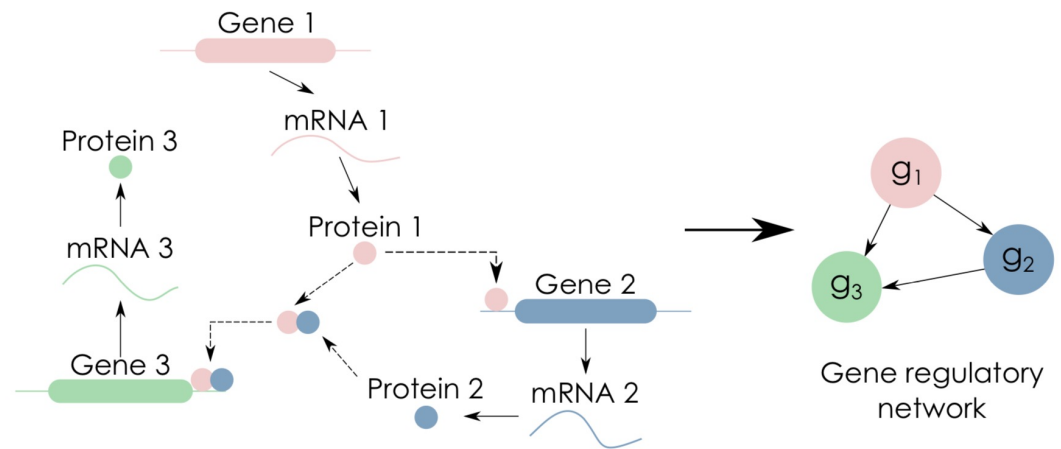
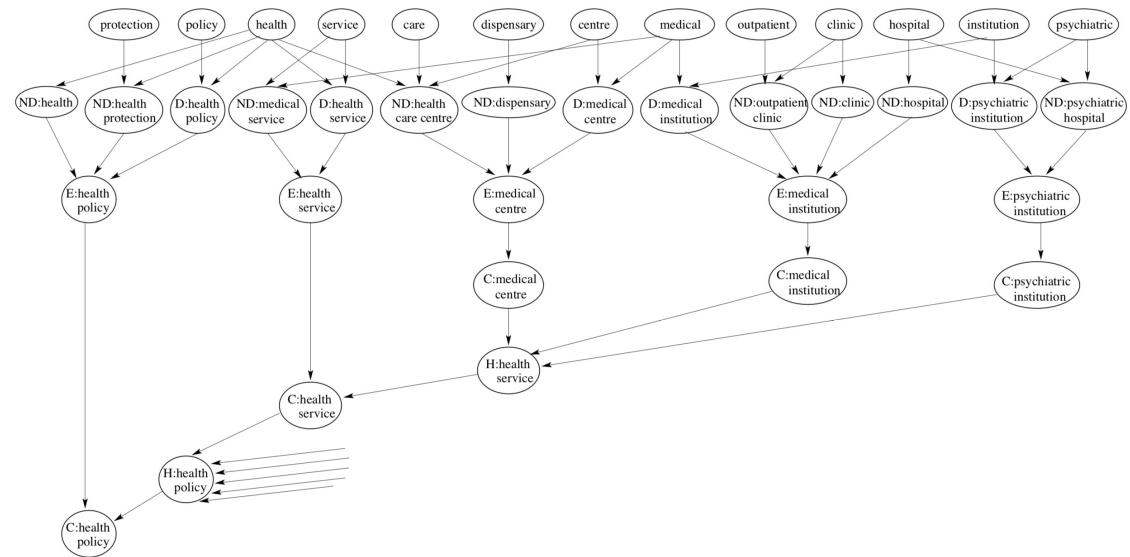


Fig. 1 A cartoon schematic of a gene regulatory network. A complex biophysical model describes the interaction between three genes, involving both direct regulation (gene 2 by gene 1) and combinatorial regulation via complex formation (gene 3 by genes 1 and 2). The abstracted structure of the system is given in the (directed) network on the right.

Document Classification



Attendance Code: 506610

Fundamental Questions in ML

- Representation
 - How to capture/model uncertainties in possible worlds?
 - How to encode our domain knowledge/assumptions/constraints?

Now we focus on this in
the last component

- Inference
 - How do I answer questions/queries according to my model and/or based on given data? e.g.: $P(X_i | \mathcal{D})$

- Learning
 - Which model is “right” for the data: e.g.: $\mathcal{M} = \arg \max_{\mathcal{M} \in \mathcal{M}} F(\mathcal{D}; \mathcal{M})$

We have discussed this quite a lot by now

A unified view of RAG and in-context learning

- **RAG**: a model improves its answers by first retrieving relevant external information and then generating a response **conditioned** on that retrieved evidence
- **In-context learning (ICL)**: a model's ability to perform a task by **conditioning** on examples or instructions provided directly in the prompt, without updating its underlying parameters.
- Both can be formulated as

$$p(y|x) = \sum_E p(y|x, E)p(E|x)$$

Where $E \sim p(E|x)$.

- For ICL, the evidence E is explicit demonstrations in the prompt.
- For RAG, the evidence E is the retrieved passages or documents



From Naïve Bayes to Bayesian Network

Attendance Code: 506610

Recap of Basic Prob. Concepts

- What is the joint probability distribution on multiple variables?

$$P(X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8)$$

- How many state configuration in total?
- Are they all needed to be represented?
- **Do we get any scientific insight?**

Recall: naïve Bayes

Parameters for Joint Distribution

- Each X_i represents outcome of tossing coin i
 - Assume coin tosses are marginally independent
 - i.e., $X_i \perp X_j$ therefore

Recall: assumption for naïve Bayes, except the latter is based on conditional independence

$$P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2) \dots P(X_n)$$

- If we use standard parameterization of the joint distribution, the independence structure is obscured and required 2^n parameters
- However, we can use a more natural set of parameters: n parameters $\theta_1, \dots, \theta_n$

Parameterization

- Example: Company is trying to hire recent graduates
- Goal is to hire intelligent employees
 - No way to test intelligence directly
 - But have access to Student's score
 - Which is informative but not fully indicative
- Two random variables
 - **I**ntelligence: $Val(I) = \{i^1, i^0\}$, high and low
 - **S**core: $Val(S) = \{s^1, s^0\}$, high and low
- Joint distribution has 4 entries
 - Need three parameters

I	S	P(I,S)
i^0	s^0	0.665
i^0	s^1	0.035
i^1	s^0	0.06
i^1	s^1	0.24

Joint distribution

Alternative Representation: Conditional Parameterization

$$P(I, S) = P(I)P(S|I)$$

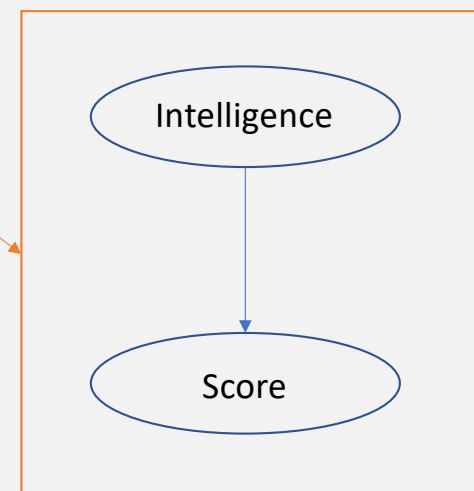
- Representation more compatible with causality
 - Intelligence influenced by Genetics, upbringing
 - Score influenced by Intelligence

- Note: BNs are not required to follow causality but they often do
- Need to specify $P(I)$ and $P(S|I)$

i^0	i^1
0.7	0.3

I	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

- Three binomial distributions (3 parameters) needed
 - One marginal, two conditionals $P(S|I = i^0)$ $P(S|I = i^1)$



Bayesian Networks (from two nodes to more nodes)

- One more variable G with $Val(G) = \{g^1, g^2, g^3\}$ to represents grades A, B, C, in addition to I and S
- If we have the following conditional independence, i.e., $P \models (S \perp G \mid I)$
- That is, Score and Grade are independent given Intelligence, i.e., knowing Intelligence, Score gives no information about class grade

Use of Conditional Independence

- Naïve Bayes assumes

$$P(X_1 \dots X_n | Y) = \prod_i P(X_i | Y)$$

i.e., that X_i and X_j are **conditionally independent** given Y , for all $i \neq j$

- Assertions

- From probabilistic reasoning $P(I, S, G) = P(I)P(S, G | I)$
- From assumption $P \models (S \perp G | I)$

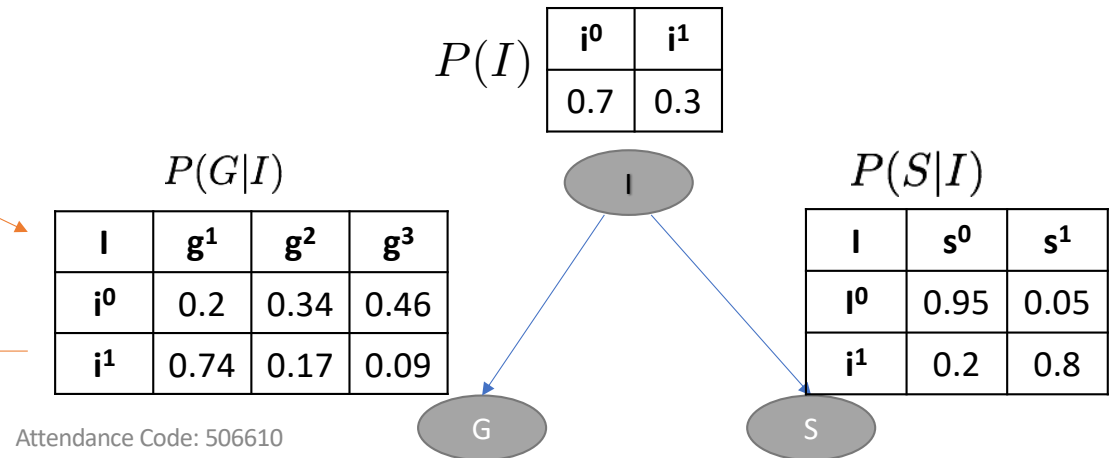
- Combining, we have

$$P(S, G | I) = P(S | I)P(G | I)$$

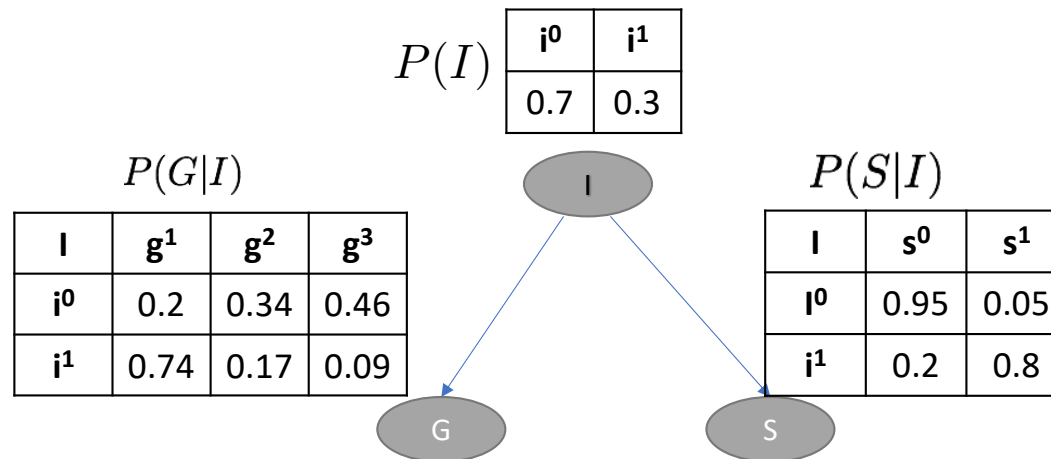
$$P(I, S, G) = P(I)P(S | I)P(G | I)$$

Two binomials, one 3-value multinomials:
7 params.

More compact than joint distribution

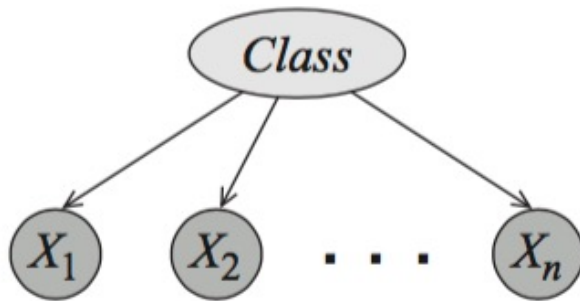
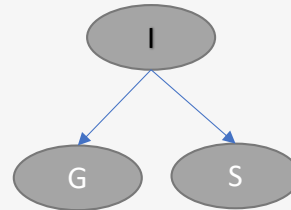


Bayesian Networks (from two nodes to more nodes)



$$\begin{aligned} P(i^1, s^1, g^2) &= P(i^1)P(s^1 | i^1)P(g^2 | i^1) \\ &= 0.3 * 0.8 * 0.17 = 0.0408 \end{aligned}$$

$$P(I, S, G) = P(I)P(S | I)P(G | I)$$



$$P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i | C)$$

Attendance Code: 506610

BN for General Naive Bayes Model

- Encoded using a very small number of parameters
- Linear in the number of variables

Bayesian Networks: Conditional Parameterization and Conditional Independences

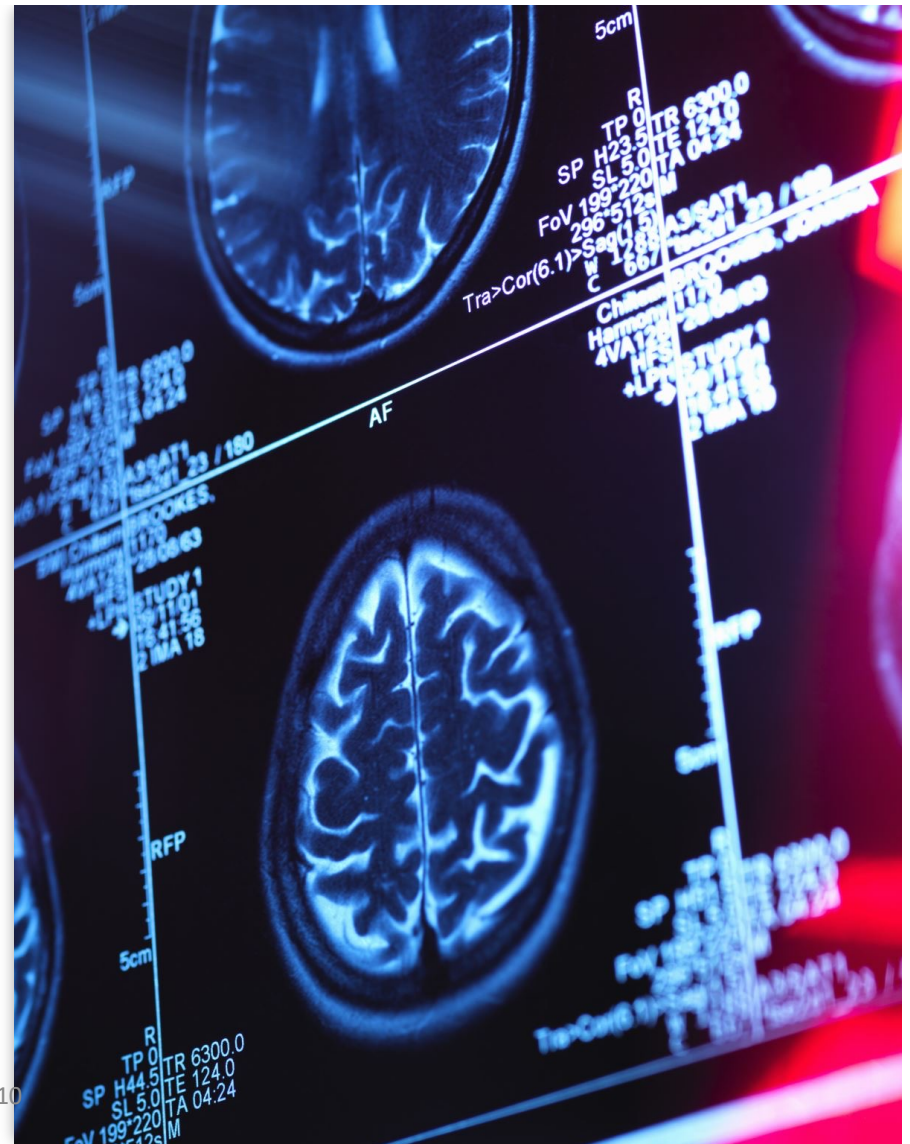
Conditional probability tables

- **Conditional Parameterization** is combined with **Conditional Independence** assumptions to produce very compact representations of high dimensional probability distributions



Application of Naive Bayes Model

- Medical Diagnosis
 - Pathfinder expert system for lymph node disease (Heckerman et.al., 1992)
- Full BN agreed with human expert 50/53 cases
- Naive Bayes agreed 47/53 cases



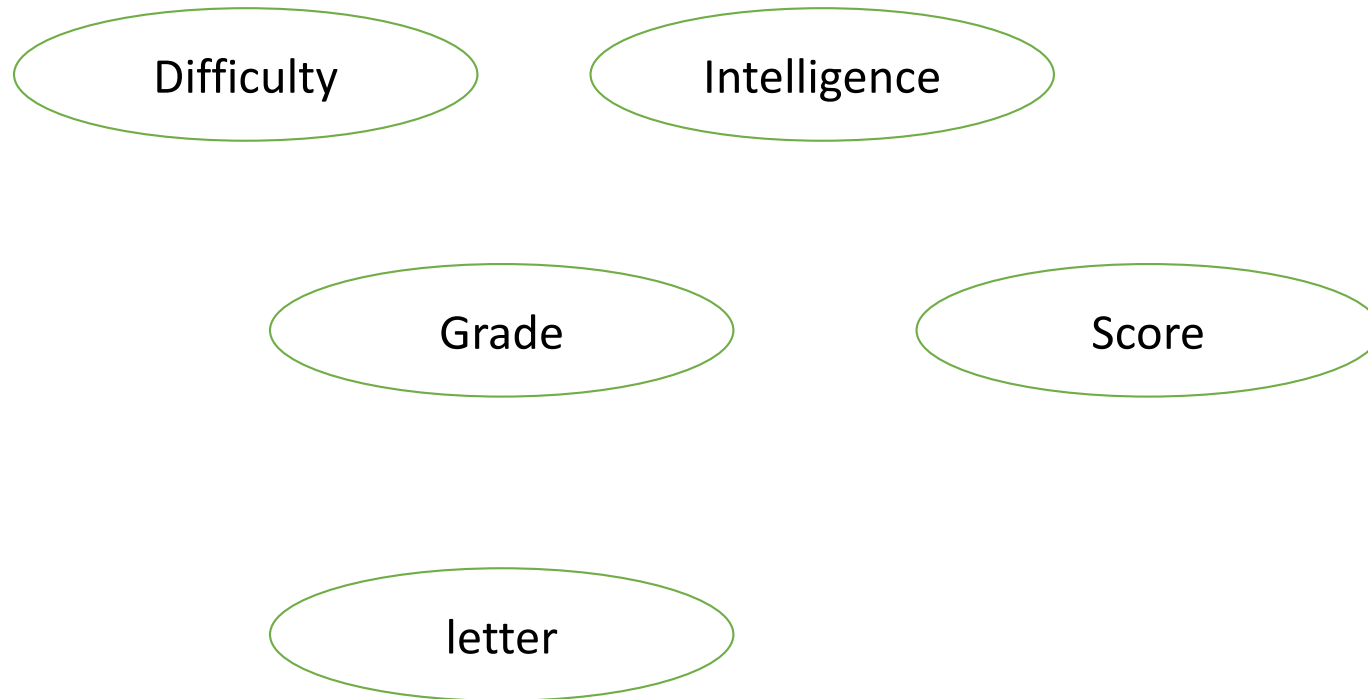
Attendance Code: 506610



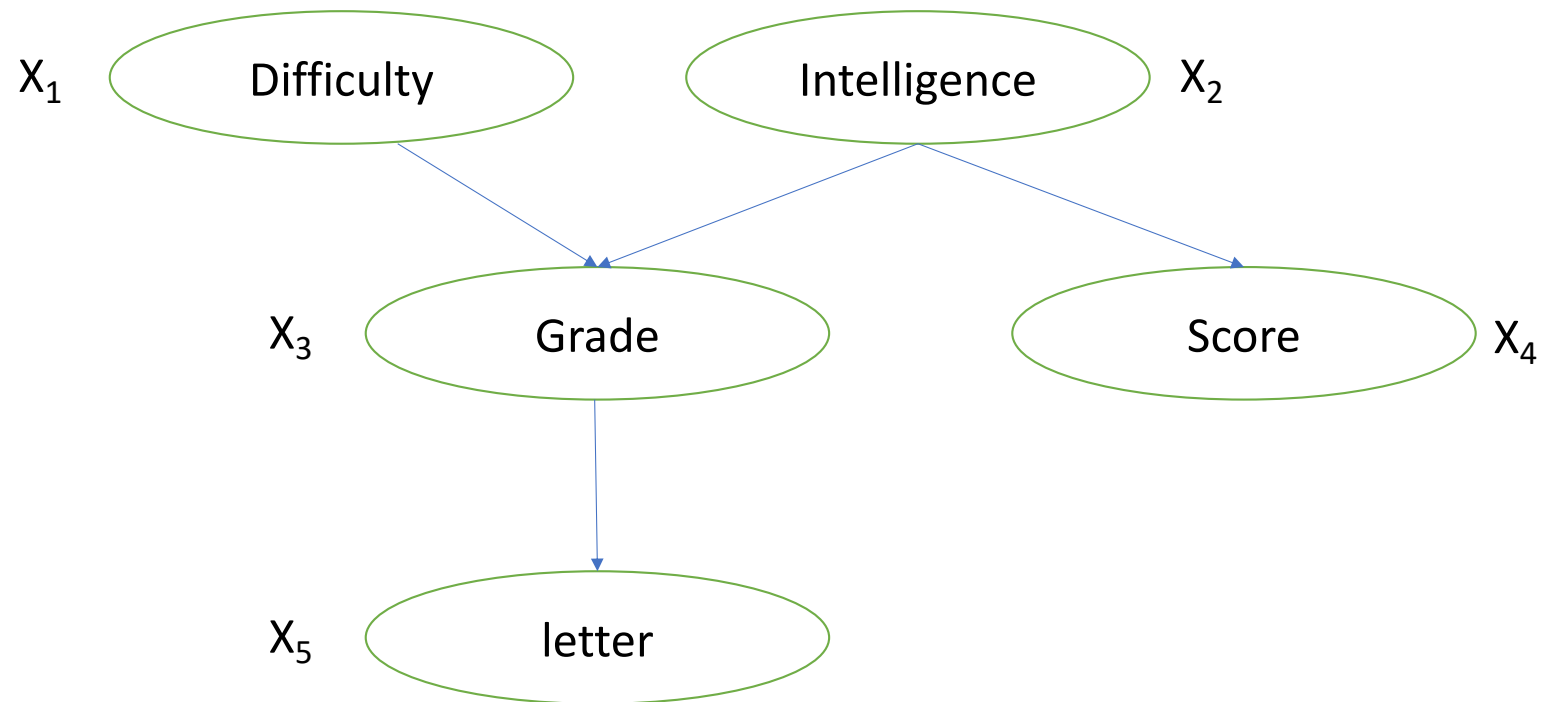
Example Bayes Networks

Attendance Code: 506610

Student Bayesian Network

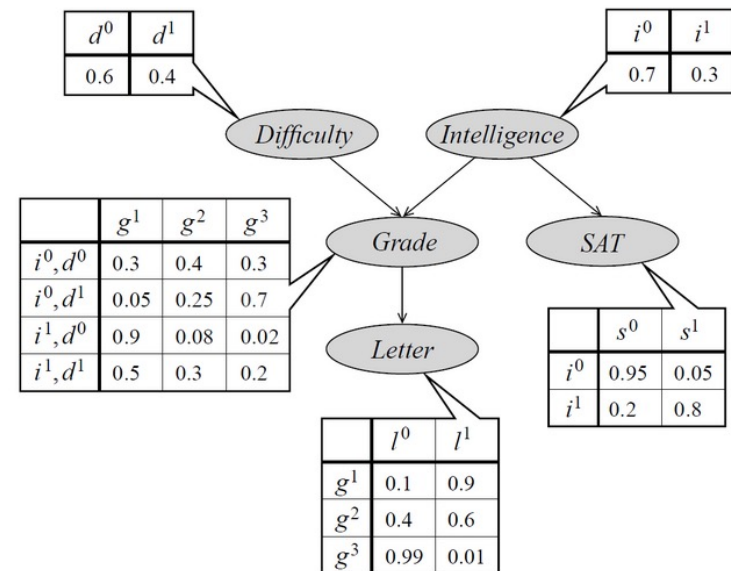


Student Bayesian Network



Student Bayesian Network

- Represents joint probability distribution over multiple variables
- BNs represent them in terms of **graphs** and conditional probability distributions (**CPDs**)
- Resulting in great savings in number of parameters needed



Joint distribution from Student BN

- CPDs: $P(X_i | pa(X_i))$

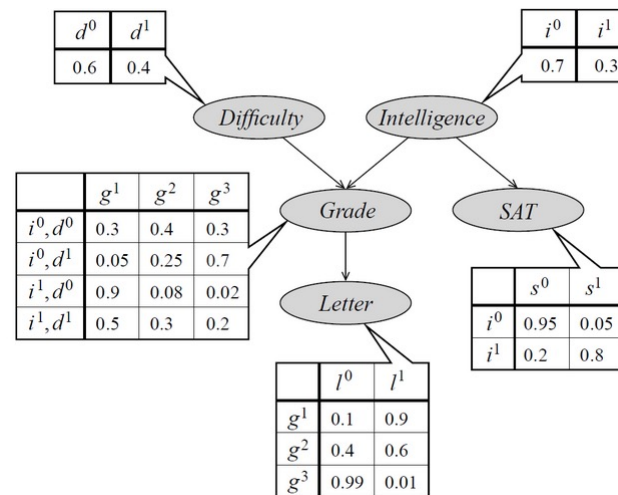
- Joint Distribution:

$$P(X) = P(X_1, X_2, \dots, X_n)$$

$$P(X) = \prod_{i=1}^n P(X_i | pa(X_i))$$

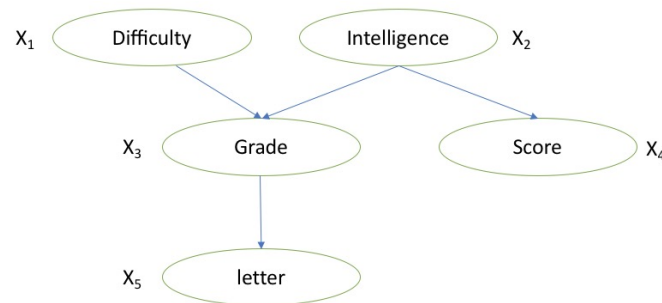
$$P(D, I, G, S, L) = P(D)P(I)P(G | D, I)P(S | I)P(L | G)$$

pa: parent nodes



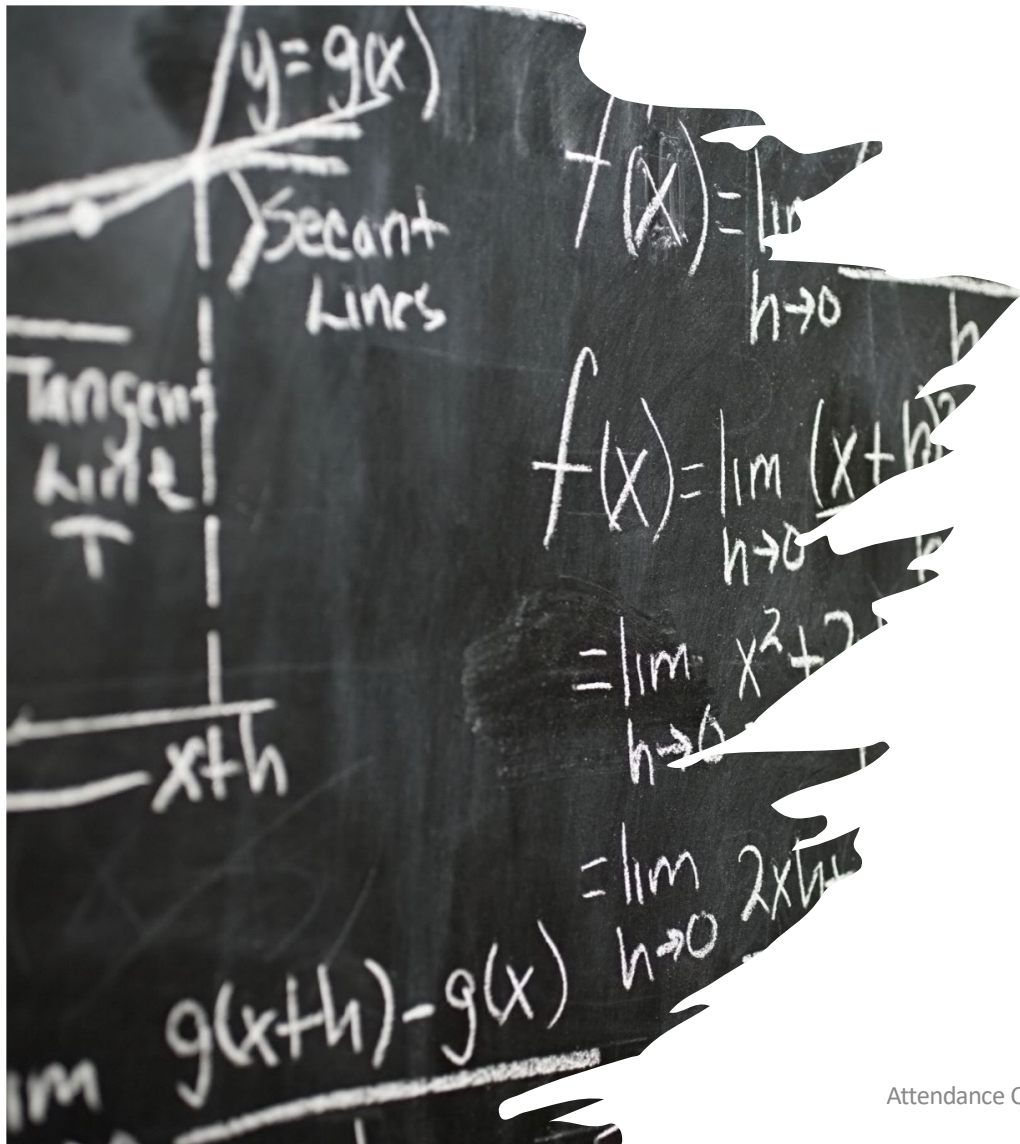
Student Bayesian Network

- If X s are conditionally independent (as described by a PGM), the joint distribution can be factored into a product of simpler terms, e.g.,



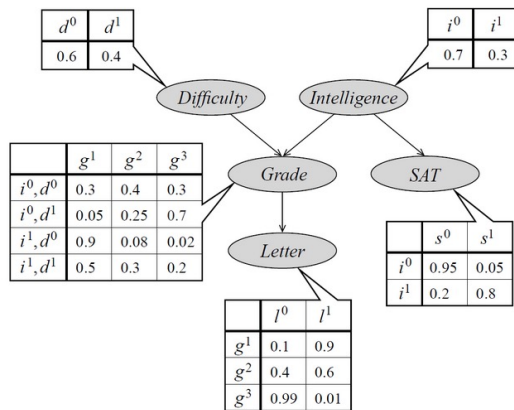
$$P(X_1, X_2, X_3, X_4, X_5) = P(X_1)P(X_2)P(X_3 | X_1, X_2)P(X_4 | X_2)P(X_5 | X_3)$$

- What's the benefit of using a PGM:
 - Incorporation of domain knowledge and causal (logical) structures
 - $1+1+7+3+3=14$, a reduction from $2^5-1 = 31$ (a rough estimation, not accurate)



Example Probability Query

Example of Probability Query



$$P(Y = y_i | E = e) = \frac{P(Y = y_i, E = e)}{P(E = e)}$$

↖
↖

Posterior Marginal
Probability of Evidence

Posterior Marginal Estimation: $P(I = i^1 | L = l^0, S = s^1) = ?$

Probability of Evidence: $P(L = l^0, S = s^1) = ?$

- Here we are asking for a specific probability rather than a full distribution

Computing the Probability of Evidence

- Probability Distribution of Evidence

$$P(L, S) = \sum_{D, I, G} P(D, I, G, L, S) \quad \text{Sum Rule of Probability}$$

$$= \sum_{D, I, G} P(D)P(I)P(G \mid D, I)P(L \mid G)P(S \mid I) \quad \text{From the Graphical Model}$$

- Probability of Evidence

$$P(L = l^0, S = s^1) = \sum_{D, I, G} P(D)P(I)P(G \mid D, I)P(L = l^0 \mid G)P(S = s^1 \mid I)$$

- More Generally $P(E = e) = \sum_{X \setminus E} \prod_{i=1}^n P(X_i \mid pa(X_i)) \big|_{E=e}$

Computing the Posterior Marginal (Method 1)

$$P(I = i^1 | L = l^0, S = s^1) = \frac{P(I = i^1, L = l^0, S = s^1)}{P(L = l^0, S = s^1)}$$

Now we know how to compute $P(L = l^0, S = s^1)$

Can you do the other one? $P(I = i^1, L = l^0, S = s^1)$

Alternatively, Rational Statistical Inference

The Bayes Theorem:

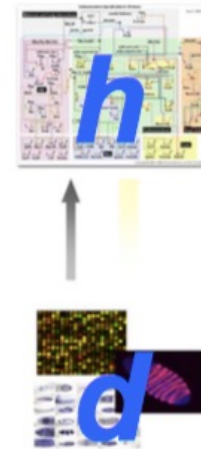
Posterior probability

Likelihood

Prior probability

$$p(h | d) = \frac{p(d | h) p(h)}{\sum_{h' \in H} p(d | h') p(h')}$$

Sum over space of hypotheses



Rational Statistical Inference (Method 2)

$$P(I = i^1 | L = l^0, S = s^1) = \frac{P(L = l^0, S = s^1 | I = i^1)P(I = i^1)}{\sum_{i \in \{i^0, i^1\}} P(L = l^0, S = s^1 | I = i)P(I = i)}$$

If we know that $P \models L \perp S | I$

$$P(I = i^1 | L = l^0, S = s^1) = \frac{P(L = l^0 | I = i^1)P(S = s^1 | I = i^1)P(I = i^1)}{\sum_{i \in \{i^0, i^1\}} P(L = l^0 | I = i)P(S = s^1 | I = i)P(I = i)}$$



What is a Graphical Model?

Attendance Code: 506610

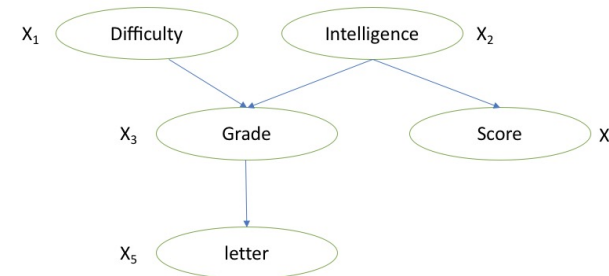
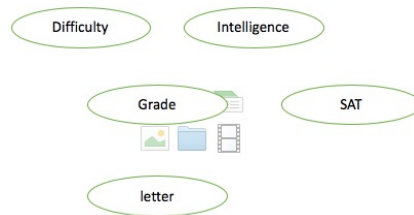
So What is a Graphical Model?

- In a nutshell,

GM = Multivariate Statistics + Structure

What is a Graphical Model?

- The informal blurb:
 - It is a smart way to write/specify/compose/design **exponentially-large probability distributions without paying an exponential cost**, and at the same time endow the distributions with **structured semantics**



- A more formal description:
 - It refers to a family of distributions on a set of random variables that are compatible with all the probabilistic independence propositions encoded by a graph that connects these variables

Two types of GMs

- Directed edges give causality relationships (**Bayesian Network** or **Directed Graphical Model**):
- Undirected edges simply give correlations between variables (**Markov Random Field** or **Undirected Graphical model**):

Bayesian Network vs. Bayesian Neural Network

- Bayesian network is the probabilistic graphical model we discuss here.
- Bayesian neural network is a neural network with Bayesian assumption on its weights.

Attendance Code: 506610

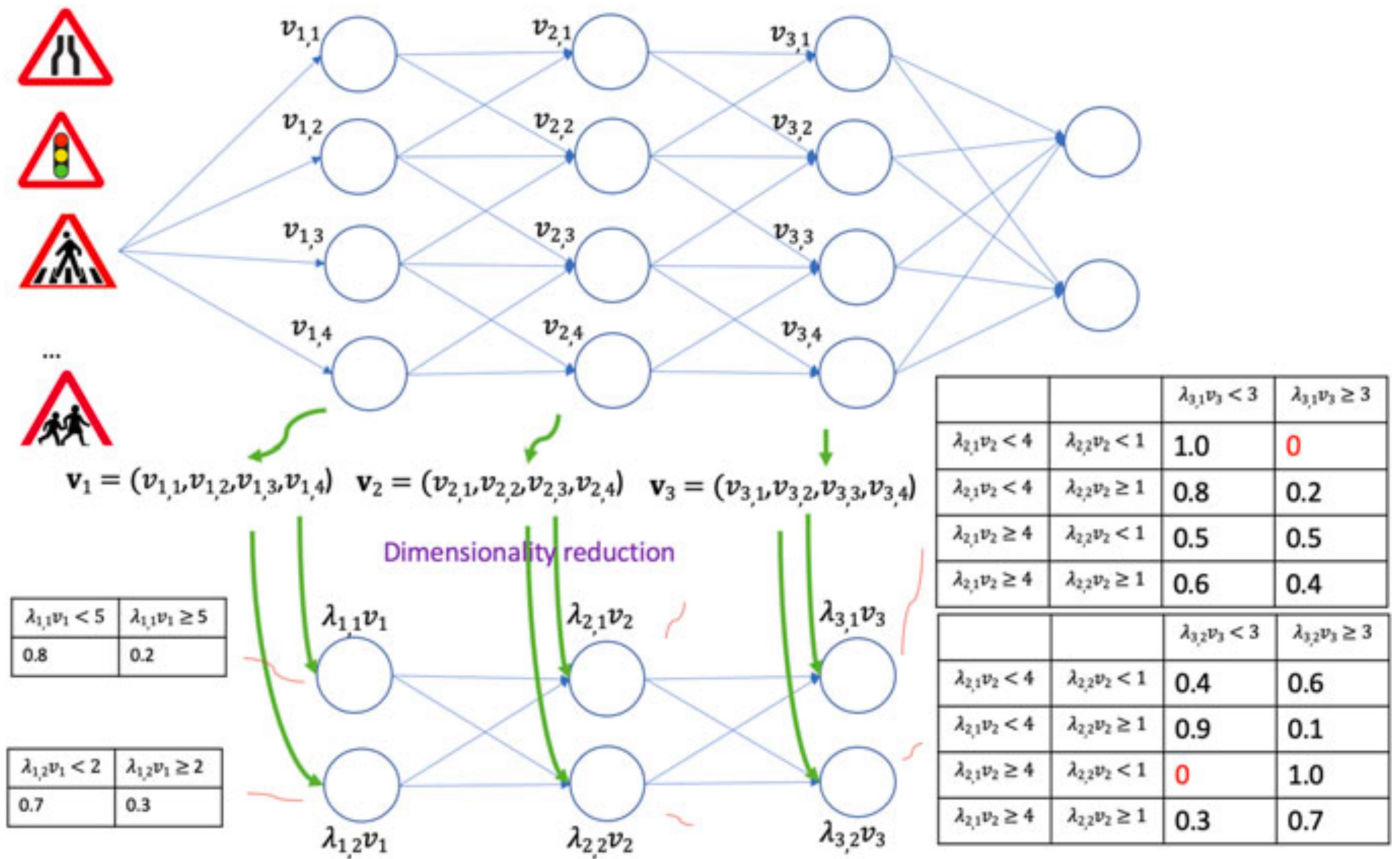




Abstraction of Neural Network as Probabilistic Graphical Model

Attendance Code: 506610

Abstraction of Neural Network as Probabilistic Graphical Model



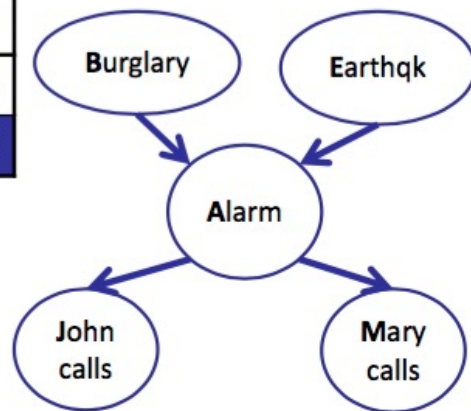


Further Example: Alarm Network

Attendance Code: 506610

Example: Alarm Network

B	P(B)
+b	0.001
-b	0.999



E	P(E)
+e	0.002
-e	0.998

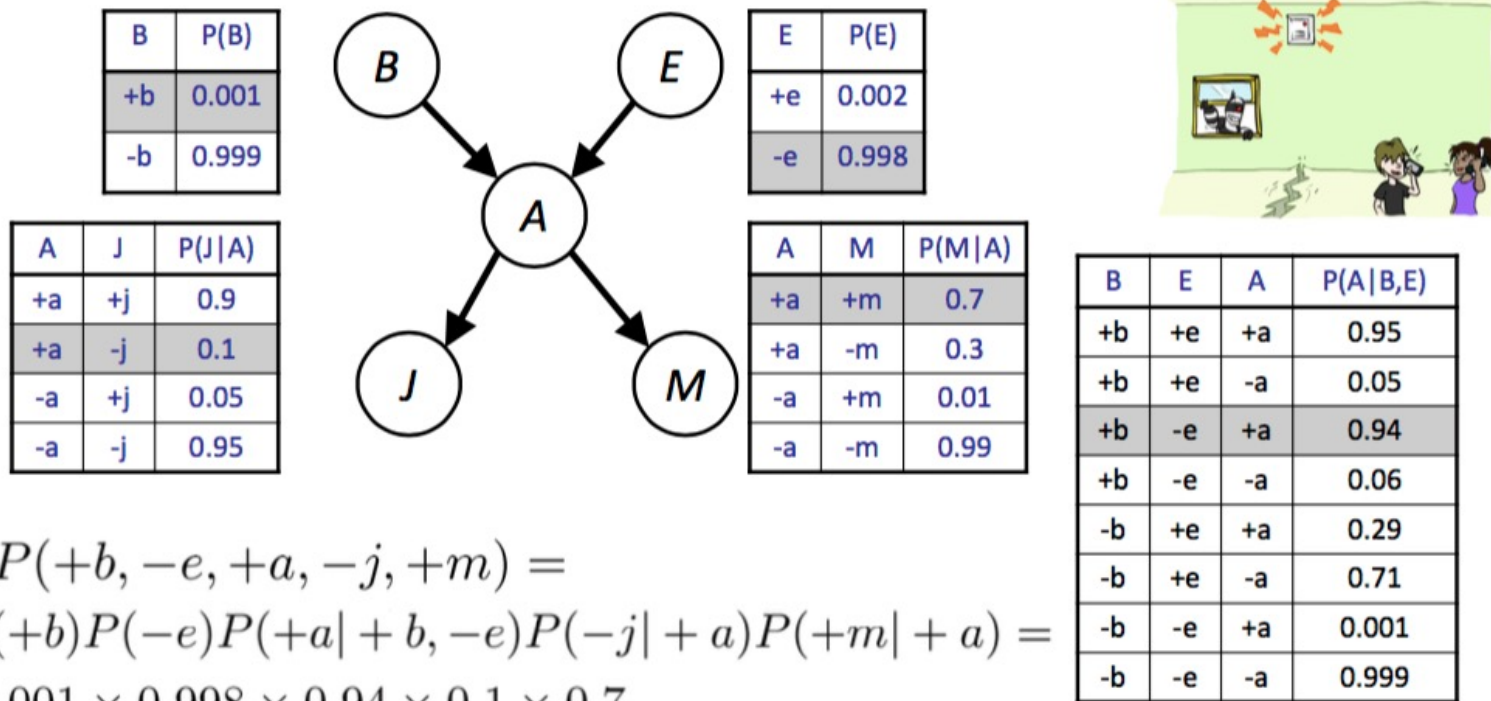


A	J	P(J A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

Example: Alarm Network



$$\begin{aligned}
 &P(+b, -e, +a, -j, +m) = \\
 &P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) = \\
 &0.001 \times 0.998 \times 0.94 \times 0.1 \times 0.7
 \end{aligned}$$